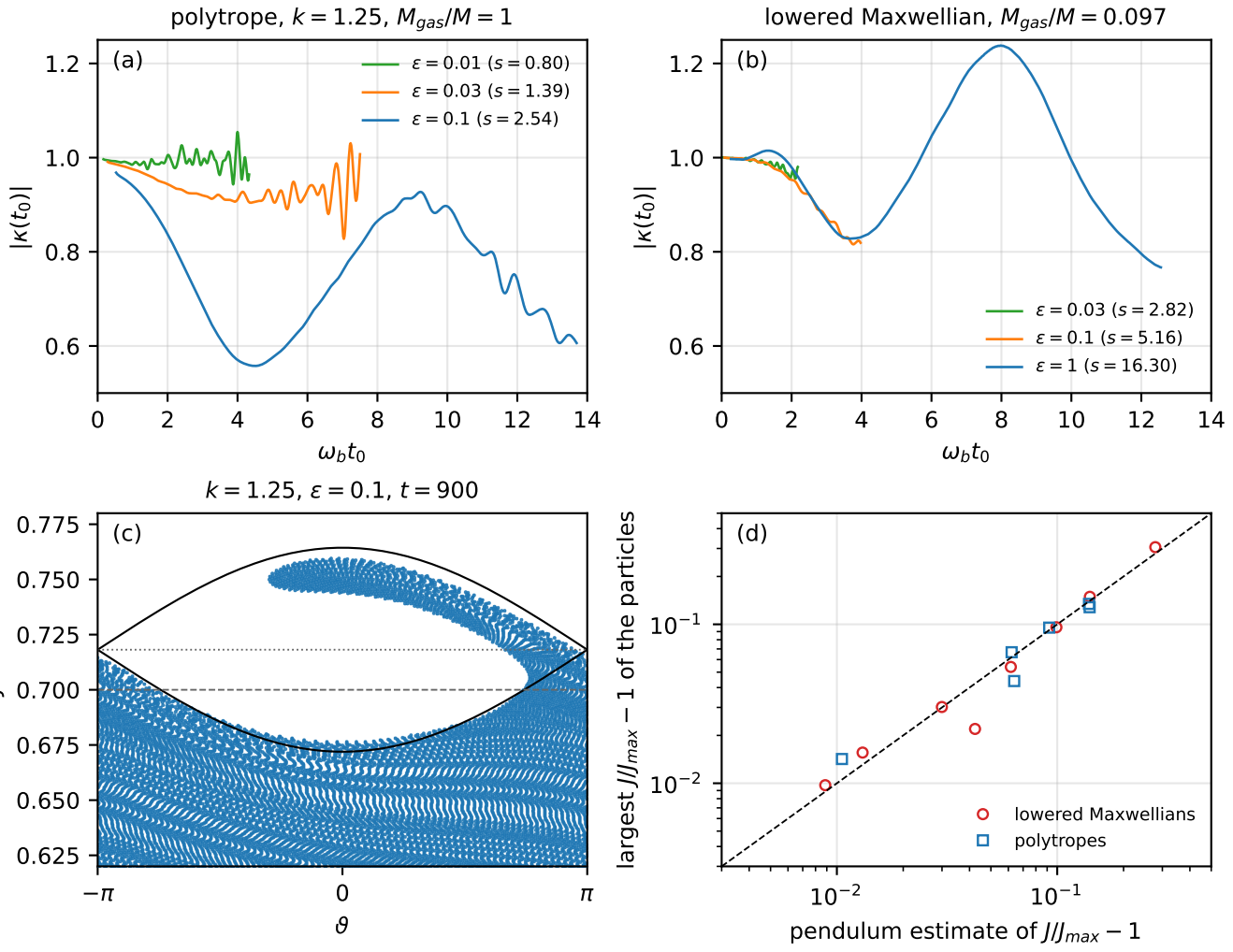

polytrope, $k = 1.25$, $M_{gas}/M = 1$
(a)
$\varepsilon = 0.01$ ($s = 0.80$)
$\varepsilon = 0.03$ ($s = 1.39$)
$\varepsilon = 0.1$ ($s = 2.54$)
$|\kappa(t_0)|$
$\omega_b t_0$
lowered Maxwellian, $M_{gas}/M = 0.097$
(b)
$\varepsilon = 0.03$ ($s = 2.82$)
$\varepsilon = 0.1$ ($s = 5.16$)
$\varepsilon = 1$ ($s = 16.30$)
$|\kappa(t_0)|$
$\omega_b t_0$
$k = 1.25$, $\varepsilon = 0.1$, $t = 900$
(c)
J
ϑ
(d)
largest $J/J_{max} - 1$ of the particles
pendulum estimate of $J/J_{max} - 1$
lowered Maxwellians
polytropes